Introduction to Control System Theory

Control Theory

Control theory is a branch of engineering and mathematics that deals with the **behavior of dynamical systems**. It focuses on designing systems that can control the behavior of other systems to produce specific outputs.

Control theory develops **mathematical models** and **algorithms** to manipulate the behavior of systems, ensuring **they operate as desired.**

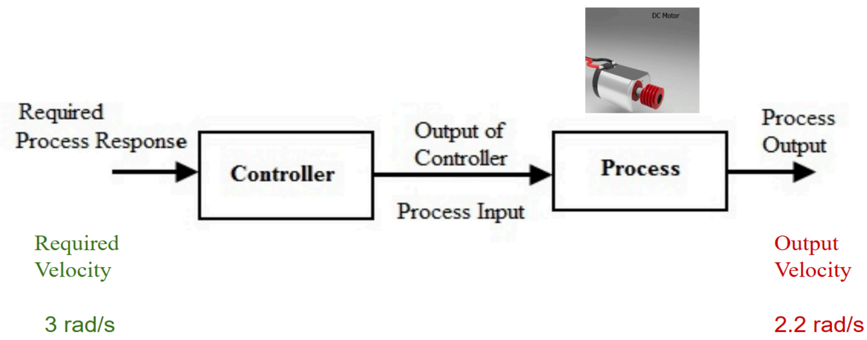
***Open-loop control:

An open-loop control system is a type of control system in which the output is not fed back to the input for correction. In other words, the system **does not use feedback** to determine if the output has achieved the desired goal. Instead, the system operates based on a predetermined sequence of actions or a fixed set of rules.

An example of an open-loop control system is a **washing machine**. The user selects the desired wash cycle and the machine operates based on a predetermined sequence of actions (filling with water, agitating, draining, spinning, etc.) without any feedback on whether the clothes are actually clean.

Open-loop control systems are generally **simpler and less expensive** than closed-loop control systems but they can be **less accurate and less able to adapt to changing conditions**.

Open Loop Control

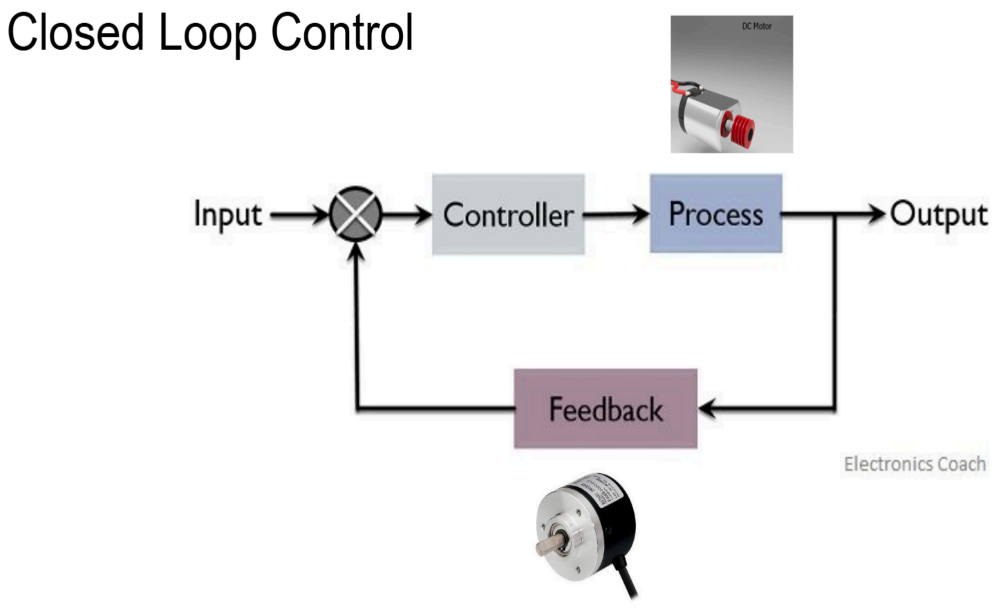


Closed loop systems:

A closed-loop control system is a type of control system in which the output is **fed back** to the input for correction. This feedback allows the system to adjust its actions based on the difference between the desired output and the actual output. The goal of a closed-loop control system is to **minimize the error** between the desired and actual outputs.

An example of a closed-loop control system is a **thermostat.** The thermostat measures the temperature in a room and compares it to the desired temperature. If the room is too cold, the thermostat sends a signal to turn on the heating system. As the room warms up, the thermostat continues to measure the temperature and adjust the heating system until the desired temperature is reached.

Closed-loop control systems are generally **more accurate and more able to adapt to changing conditions** than open-loop control systems but they can be **more complex and more expensive.**



***Characteristics of Feedback Control System:

A feedback system is a type of control system that uses feedback to adjust its actions based on the difference between the desired output and the actual output. Some characteristics of a feedback system include:

Power amplification: A feedback system can use a small amount of power to control a large amount of power. For example, a small electrical signal can be used to control a large motor.

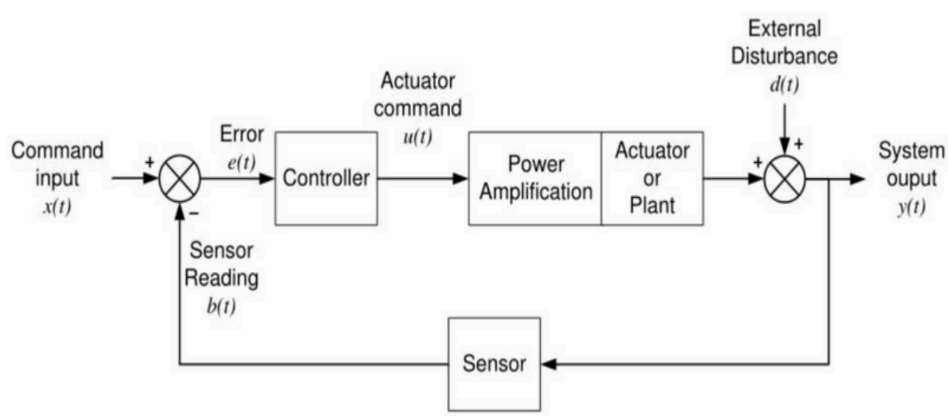
Actuator: An actuator is a component of a feedback system that converts the control signal into an action. For example, a motor is an actuator that converts an electrical signal into mechanical motion.

Feedback: Feedback is the process of measuring the output of a system and comparing it to the desired output. The difference between the desired and actual output is used to adjust the system’s actions.

Error signal: The error signal is the difference between the desired and actual outputs. It represents the amount by which the system needs to be adjusted in order to achieve the desired output.

Controller: The controller is the component of a feedback system that processes the error signal and generates a control signal to adjust the system’s actions.

Feedback Diagram



Components:

- **Input:** The signal or commands applied to the system.
- **Output**: The response of the system to the input signals.
- **Controller**: The component responsible for generating control signals based on the system's output and desired behavior.

- **Feedback**: Information about the system's output that is fed back to the controller to adjust the control signals.

Goal of a control system:

The goal of a control system is to achieve a desired output by manipulating the inputs to a system. Some common goals of control systems include:

Regulation: Regulation is the process of maintaining a system's output at a constant value or within a desired range. For example, a thermostat regulates the temperature in a room by turning the heating or cooling system on and off as needed.

Tracking: Tracking is the process of making a system's output follow a desired reference signal. For example, a cruise control system tracks the desired speed of a car by adjusting the throttle.

Optimization: Optimization is the process of finding the best possible solution to a problem. In control systems, optimization can involve finding the best set of inputs to achieve a desired output while minimizing costs or maximizing performance.

Characteristics of a good control system:

- **Accurate**: It must reach the desired goal with no final error. Integral control is often used to eliminate this "steady-state error".
- **Fast**: It should reach the goal quickly. This is known as having a fast

transient response, which can be improved with derivative control.

- **Stable and Well-Damped**: The system's response must remain bounded and not grow without limit. It should approach the target with minimal oscillations or overshoot.
- **Quick to Settle**: The system should have a short **stability time**, meaning it transitions rapidly from its transient phase to its final, steady state.

Feedback: Feedback involves using information about the output of a system to adjust the control input, thereby regulating the system's behavior.
Ex. The thermostat measures the current room temperature(feedback), compares it to the desired temperature set by the user, adjusts the heating or cooling.

Stability: Stability ensures that the system's response remains bounded over time, even in the presence of disturbances or changes in operating conditions.
Ex. balancing a pencil on the tip of a finger. Moving your finger slightly, the pencil wobbles but eventually returns to its position. The pencil returns to equilibrium despite disturbances illustrates stability.

Transient State:

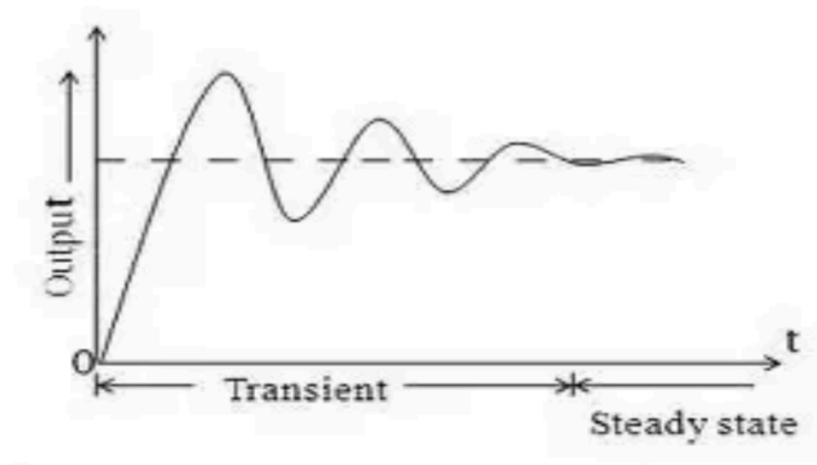
The transient response, on the other hand, refers to the behavior of the system during the initial period after a change in the input or a disturbance. This is the short-term behavior of the system as it responds to changes and moves towards its new steady-state operating point.
Ex. Turning on a light switch, the time it takes for the light to reach its steady state and any oscillations or overshoot during this transition represent the transient

response of the lighting system.

Steady-State:

In the context of a control system, the steady-state response refers to the behavior of the system after it has settled and reached a stable operating point. This is the long-term behavior of the system after any initial transients have died out.

Ex. In an elevator system, when you press a floor button, the elevator initially accelerates, then moves at a constant speed, and finally decelerates as it reaches the desired floor.

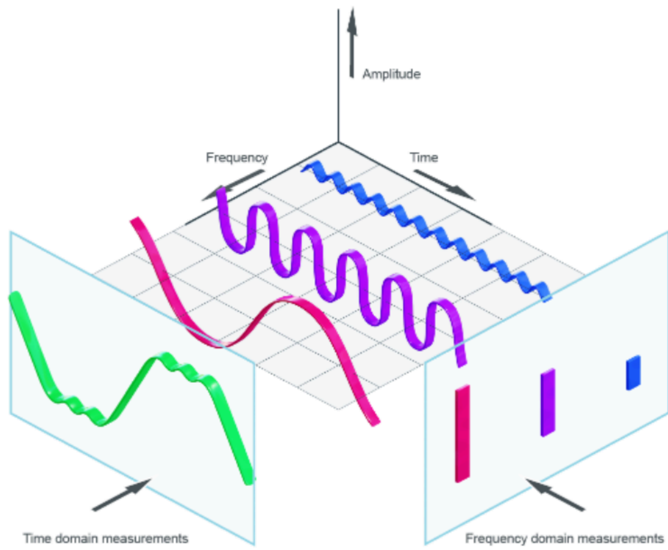


Why do we need the conversion of signal from time domain to frequency domain?

-One technical reason for converting a time-domain signal to a frequency-domain signal is that it can simplify the analysis of signals and systems. Many mathematical operations that are difficult to perform in the time domain become much simpler in the frequency domain. For example, convolution in the time domain becomes multiplication in the frequency domain. This can make it easier to analyze how a system will respond to different inputs.

Another technical reason is that many systems and devices respond differently to different frequencies. By converting a time-domain signal to a frequency-domain signal, you can see how the system or device will respond to each frequency component of the input signal. This can make it easier to design filters or other systems that manipulate the frequency content of a signal.

Time Domain to Frequency Domain



Transfer Function

A **transfer function** is a mathematical model that represents the relationship between the input and output of a system. Formally, it's defined as the ratio of the Laplace Transform (LT) of the output, Y(s), to the Laplace Transform of the input, X(s), when all initial conditions are assumed to be zero.

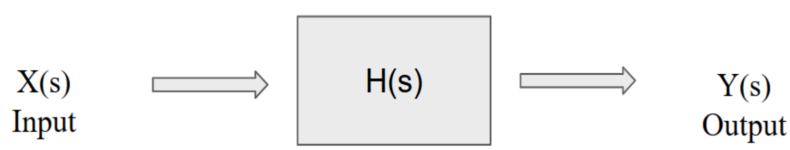
H(s)=Y(s)/X(s)

This function provides a powerful way to understand how a linear system will behave in response to different inputs. By analyzing the transfer function, you can determine the system's characteristics in both the time and frequency domains.

Specifically, the transfer function is used to:

- Determine the **stability** of a system.
- Analyze its **transient response** (how it behaves during changes) and its **steady-state response** (its long-term behavior).
- Describe how the system reacts to an impulse, known as the **impulse response**.
- Design **controllers** that modify the system's behavior to achieve a desired performance.

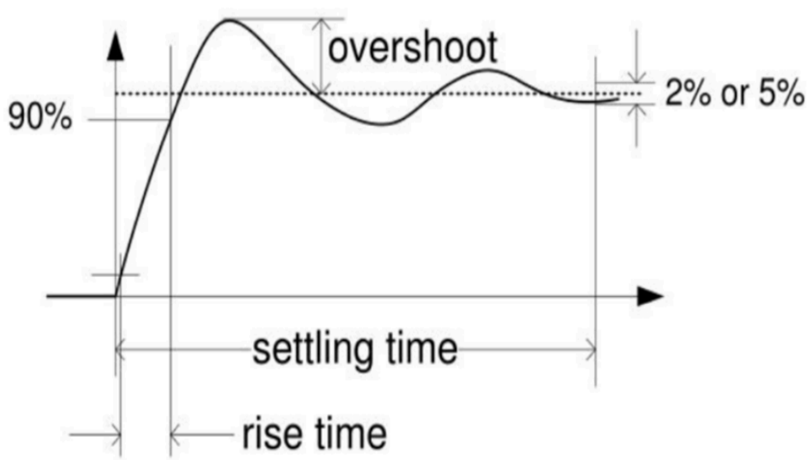
Transfer Function



Overshoot and Undershoot: Overshoot refers to the amount by which the system's output exceeds its final steady-state value in response to a change in the input or a disturbance. Undershoot is the opposite of overshoot and refers to the amount by which the system's output falls below its final steady-state value.

Rise time: It is the time taken for the system's output to rise from a specified low value to a specified high value in response to a change in the input or a disturbance. It is typically measured as the time it takes for the output to rise from 10% to 90% of its final steady-state value.

Settling time: Settling time is the time it takes for the system's output to settle within a specified tolerance band around its final steady-state value after a change in the input or a disturbance. It is typically measured as the time it takes for the output to remain within a specified percentage (usually 2% or 5%) of its final steady-state value.



Effect of Controller Function:

Proportional Control:

A proportional controller is a type of feedback controller that calculates the control action as a proportion of the error signal. The error signal is the difference between the desired setpoint and the measured process variable. The control action is calculated by multiplying the error signal by a constant gain factor. This means that The control action is directly proportional to the error signal.

$$\boxed{u(t) = Kp*e(t)}$$

Here e(t) is the error signal and u(t) is the output signal of the controller.

Proportional controllers are simple to implement and can provide good performance for many systems. However, they can also suffer from steady-state error and may not be able to completely eliminate the error between the setpoint and the process variable.

Integral Control:

Integral control is a type of feedback control that calculates the control action based on the accumulated error over time. The error signal is the difference between the desired setpoint and the measured process variable. The integral controller calculates the control action by multiplying the integral of the error signal by a constant gain factor.

$$u(t) = K_i \int_0^t e(t)dt$$

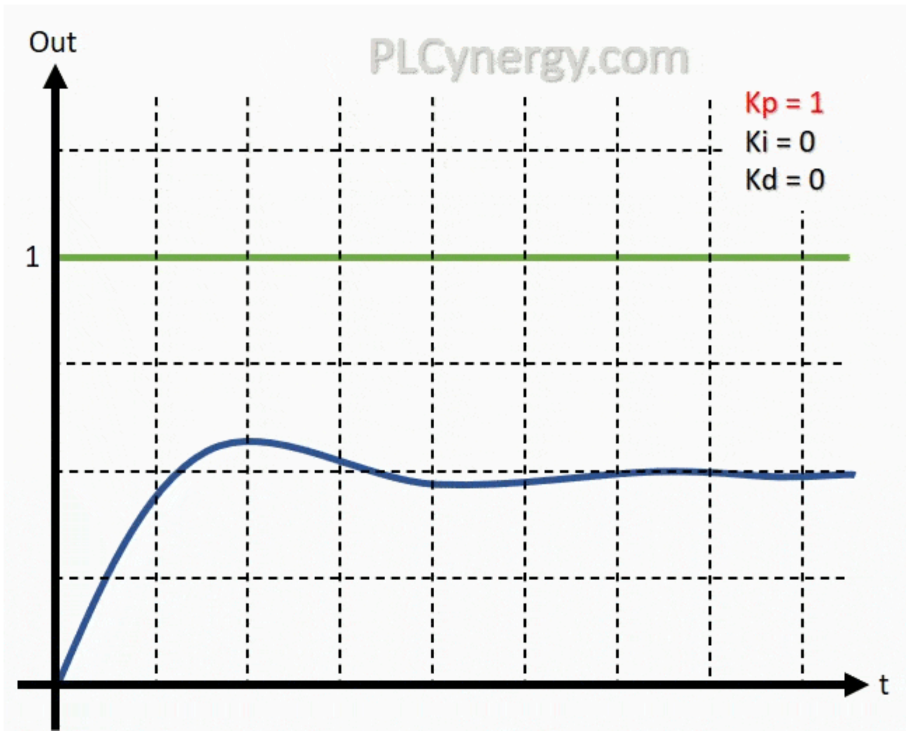
The integral action accumulates the error over time and produces a control action that is proportional to the accumulated error. This means that even small errors will eventually result in a large control action if they persist for a long time. This can help to eliminate steady-state errors and improve the accuracy of the system.

Derivative/Differential Control:

Derivative control is a type of feedback control that calculates the control action based on the rate of change of the error signal. The error signal is the difference between the desired setpoint and the measured process variable. The derivative controller calculates the control action by multiplying the derivative of the error signal by a constant gain factor.

$$u(t) = K_d \frac{d}{dt} e(t)$$

The derivative action produces a control action that is proportional to the rate of change of the error signal. This means that the controller responds to how quickly the error is changing, rather than its magnitude. This can help to improve the stability of the system and reduce overshoot.



Advantages and disadvantages of Control algorithms:

Proportional, integral, and derivative control algorithms each have their own advantages and drawbacks.

Proportional control is simple to implement and can provide good performance for many systems. However, it can suffer from steady-state error and may not be able to completely eliminate the error between the setpoint and the process variable. Additionally, a high gain can cause the system to oscillate or become unstable.

Integral control can help to eliminate steady-state errors and improve the accuracy of the system. However, it can also introduce overshoot and increase the settling time of the system. Additionally, a high gain can cause the system to oscillate or become unstable.

Derivative control can improve the stability of the system and reduce overshoot. However, it can also amplify noise in the system and make it more sensitive to measurement errors. Additionally, derivative control alone is not sufficient to control a system and must be used in combination with proportional and/or integral control.

PID Controller

$$\text{Proportion al control}: \quad u(t) = K_p e(t) \qquad \frac{U(s)}{E(s)} = K_p$$

$$\text{Integral control}: \quad u(t) = K_i \int_0^t e(t)dt \qquad \frac{U(s)}{E(s)} = \frac{K_i}{s}$$

$$\text{Differenti al control}: \quad u(t) = K_d \frac{d}{dt} e(t) \qquad \frac{U(s)}{E(s)} = K_d s$$

It produces an output, which is the combination of the outputs of proportional, integral & derivative controllers.

$$u(t) \propto e(t) + \int e(t) + \frac{d}{dt} e(t)$$

$$\gg u(t) = K_P e(t) + K_I \int e(t) + K_D \frac{d}{dt} e(t)$$

Laplace transform in both side

$$U(S) = K_P E(S) + \frac{K_I}{S} E(S) + K_D S E(S)$$

$$U(S) = E(S) \left(K_P + \frac{K_I}{S} + K_D S \right)$$

$$\frac{U(S)}{E(S)} = \left(K_P + \frac{K_I}{S} + K_D S \right) = \frac{K_P S + K_I + K_D S^2}{S}$$

Effect of PID controller function

- **Proportional Action (Kp)**
  - Simplest Controller Function

  If we increase or decrease the Kp constant, it'll try to reach the expected value.

- **Integral Action (Ki)**
  - Eliminates steady-state error
  - Can cause Oscillations

  Get the graph around the expected output, but the oscillated value is higher in this case (above expected output or below expected output).

- **Derivative Action (Kd)**
  - Effective in transient periods
  - Provides faster response (higher sensitivity)
  - Never used alone

  Kd actually rescues the oscillation and after increasing Kd, we'll get a faster response.

Effect of Controller Functions

• Proportional Action

Simplest Controller Function, The P term helps to reduce the steady-state error and improve the system's responsiveness. However, too high of a proportional gain can lead to instability and oscillations in the system's response.

• Integral Action

Eliminates steady-state error, The I term helps to improve the system's stability and eliminate any bias in the system. However, too high of an integral gain can lead to overshoot and instability in the system's response.

• Derivative Action ("rate control")

Effective in transient periods, The D term helps to improve the system's stability and reduce the effects of disturbances in the system. However, too high of a derivative gain can lead to noise amplification and instability in the system's response.

PID Tuning

How to get the PID parameter values ?

- If we know the transfer function, analytical methods can be used (e.g., root-locus method) to meet the transient and steady-state specs.
- When the system dynamics are not precisely known, we must resort to experimental approaches.

Ziegler-Nichols Rules for Tuning PID Controller

Using only Proportional control, turn up the gain until the system oscillates without dying down, i.e., is marginally stable. Assume that K and P are the resulting gain and oscillation period, respectively.

Then Use	for P control	for PI control	for PID control	
	$K_p = 0.5\,K$	$K_p = 0.45\,K$	$K_p = 0.6\,K$	Ziegler-Nichols Tuning for second or higher order systems
		$K_i = 1.2 / P$	$K_i = 2.0 / P$	
			$K_d = P / 8.0$	

Generating PID parameters

- **Auto-Tuning Algorithms**:
    - Relay-based Auto-Tuning
    - A relay is introduced in the control loop.
    - Observes the system's oscillatory response.
    - Calculates PID parameters based on oscillation characteristics.
- **Trial and Error:** Iteratively adjust parameters based on system response.

Advantages and Applications of PID control

Advantages:

- Simple: easy to understand.
- Effective: Accurate and stable, even in dynamic environments.
- Robust: Can be adopted to different robot systems

Application:

- Arm positioning
- Robot Navigation
- Speed Control
- Balance control